TRANSFORMATIONS · DICTIONARY

Transformation ↔ matrix

Once a basis is fixed the two languages are interchangeable, and every statement about a map has a matrix statement beside it. The point of reading them in parallel is to notice that neither side is primary — the matrix is not an approximation to the transformation, it is the transformation written in coordinates.

7
ENTRIES

THE CORRESPONDENCE ITSELF 3

1 **The map and its matrix**

unique $m \times n$ matrix per linear T

$$T(\mathbf{x}) = A\mathbf{x}$$

Every linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$ is matrix multiplication and nothing else — there is no more general kind of linear map hiding elsewhere. The uniqueness matters as much as the existence: two different matrices cannot represent the same map in the same basis.

2 **Columns are images of basis vectors**

how the matrix is built

$T(\mathbf{e}_j) = j$-th column of A

The whole construction in one line. Feed the map each basis vector in turn and record what comes back — the results are the columns. This is why knowing a linear map on a basis is knowing it everywhere.

3 **Dimensions**

domain $\mathbb{R}^n$, codomain $\mathbb{R}^m$

n columns, m rows

Columns count inputs, rows count outputs. A linear operator has $V = W$, so its matrix is square — which is the only case where eigenvalues, determinants and inverses are even defined.

OPERATIONS TRANSLATE 2

4 **Composition**

note the order

$$S \circ T \leftrightarrow BA$$

Applying T then S corresponds to the product BA, not AB — the matrix acting first stands on the right, because it meets the vector first. This is the reason matrix multiplication is defined the way it is rather than entry by entry.

5 **Identity and inverse**

T bijective $\iff$ A invertible

$$I \leftrightarrow I_n,\ T^{-1} \leftrightarrow A^{-1}$$

The identity transformation is the identity matrix in every basis — a change of basis cannot alter it (nor any scalar multiple cI). Invertibility transfers exactly, which is what makes the invertibility equivalence a statement about transformations too.

SUBSPACES TRANSLATE 2

6 **Image and kernel**

always

$$\operatorname{im}(T) \leftrightarrow \operatorname{Col}(A),\ \ker(T) \leftrightarrow \operatorname{Null}(A)$$

The reachable outputs are the column space; the collapsed inputs are the null space. So rank-nullity is a statement about transformations that happens to be provable by row reduction.

7 **Isomorphism**

$\dim V = \dim W$, both finite

invertible square matrix

Two spaces of the same finite dimension are isomorphic, and the isomorphism is a change of coordinates. This is why $\mathcal{P}_3$ and $\mathbb{R}^{2\times 2}$ are the same object twice — both four-dimensional, so both are $\mathbb{R}^4$ in disguise.

The dictionary is exact and it is one-to-one, but only once a basis is chosen. Change the basis and the same transformation generally gets a different matrix — which is what makes similarity the right notion of sameness for matrices, and why properties surviving a basis change are the ones worth naming.